



Tour Into the Picture

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Motivation: How to reconstruct 3D scenes from a single 2D picture or photograph?

1 Image Selection

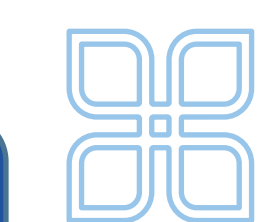
2 Foreground Object Specification

- Pixel coordinates of 4 vertices
- Create mask & Extract foreground image



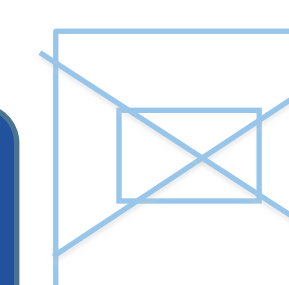
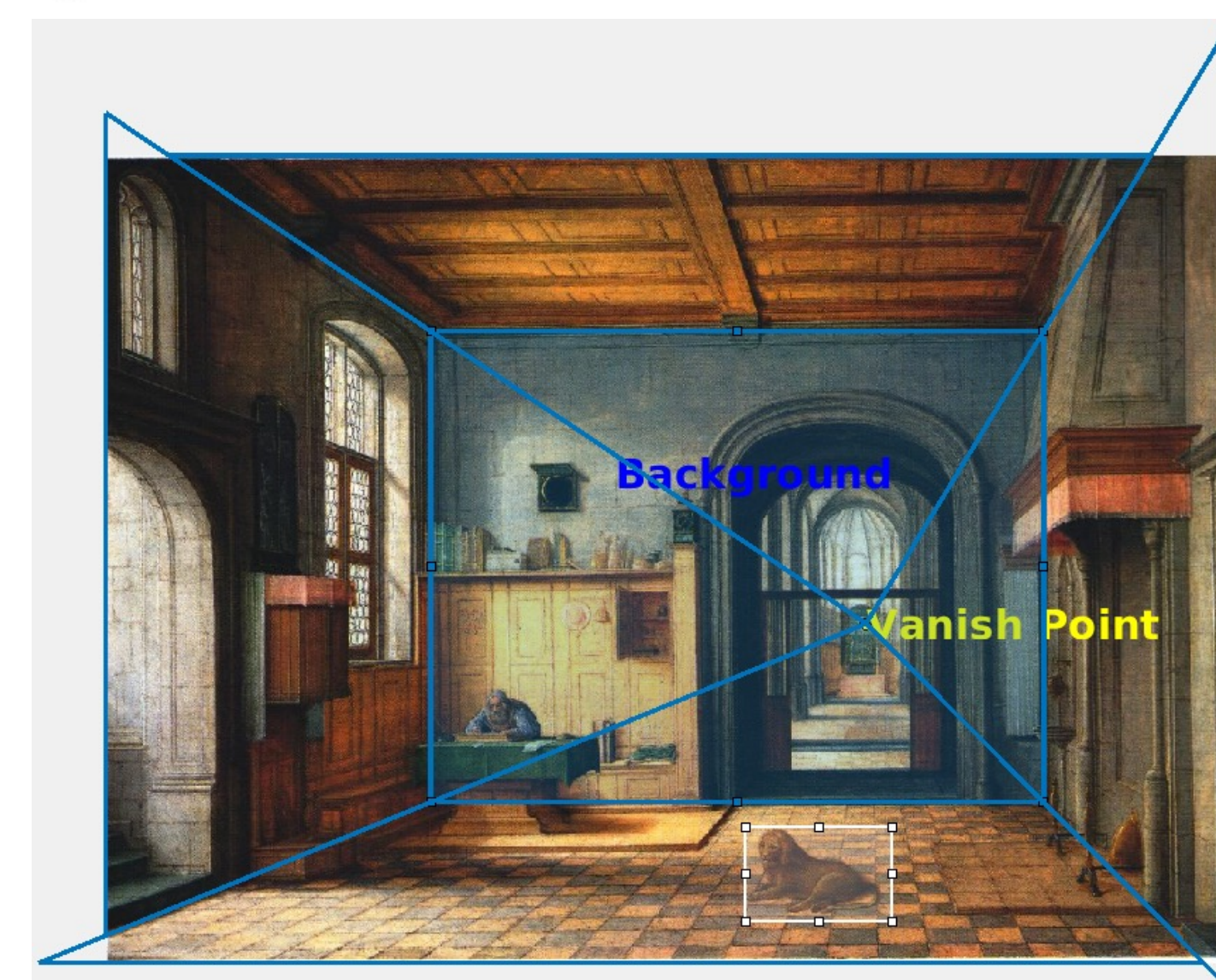
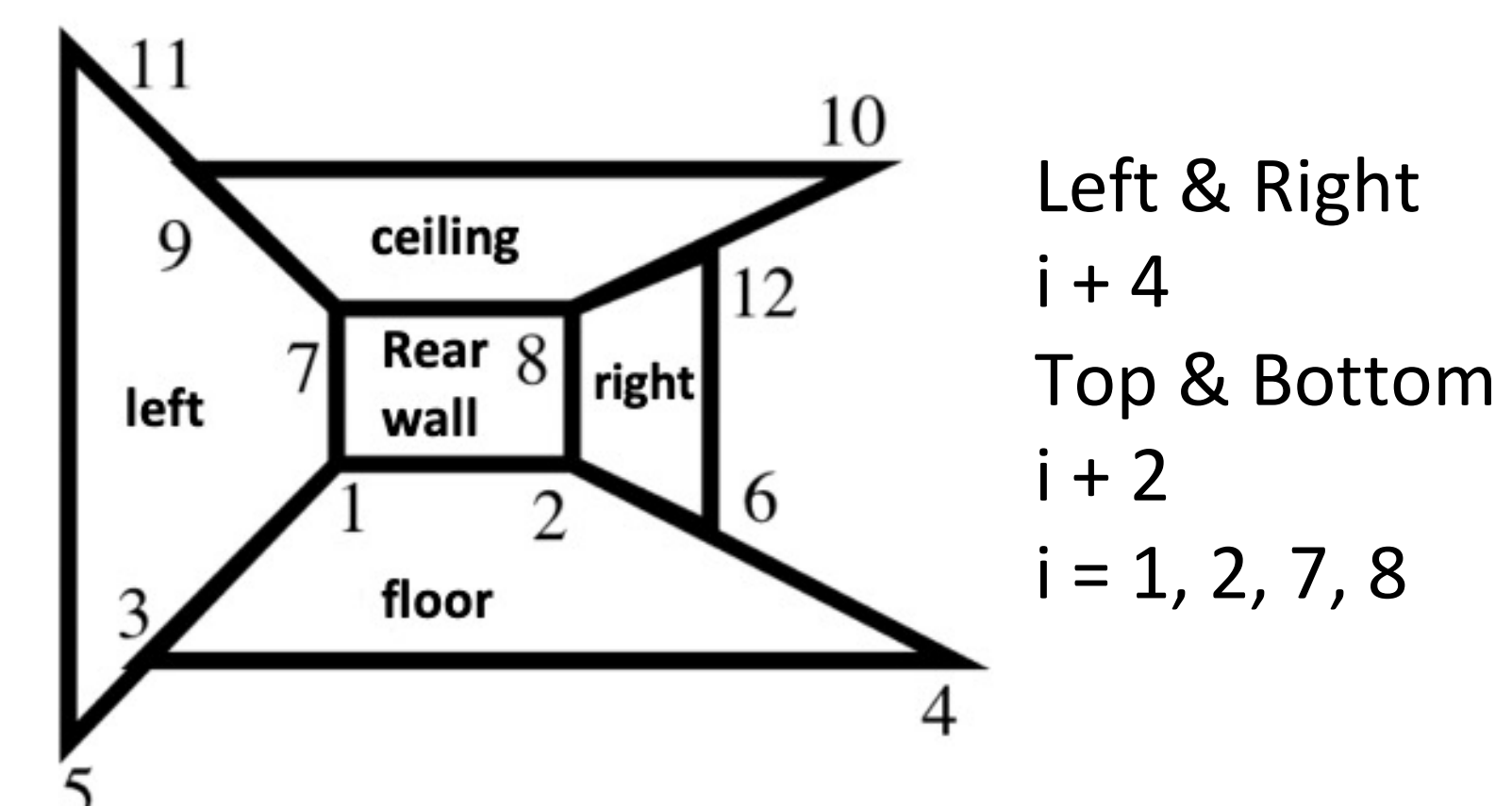
3 Background Selection

- Vanishing Point & Inner Rectangle
- Calculation of background vertices



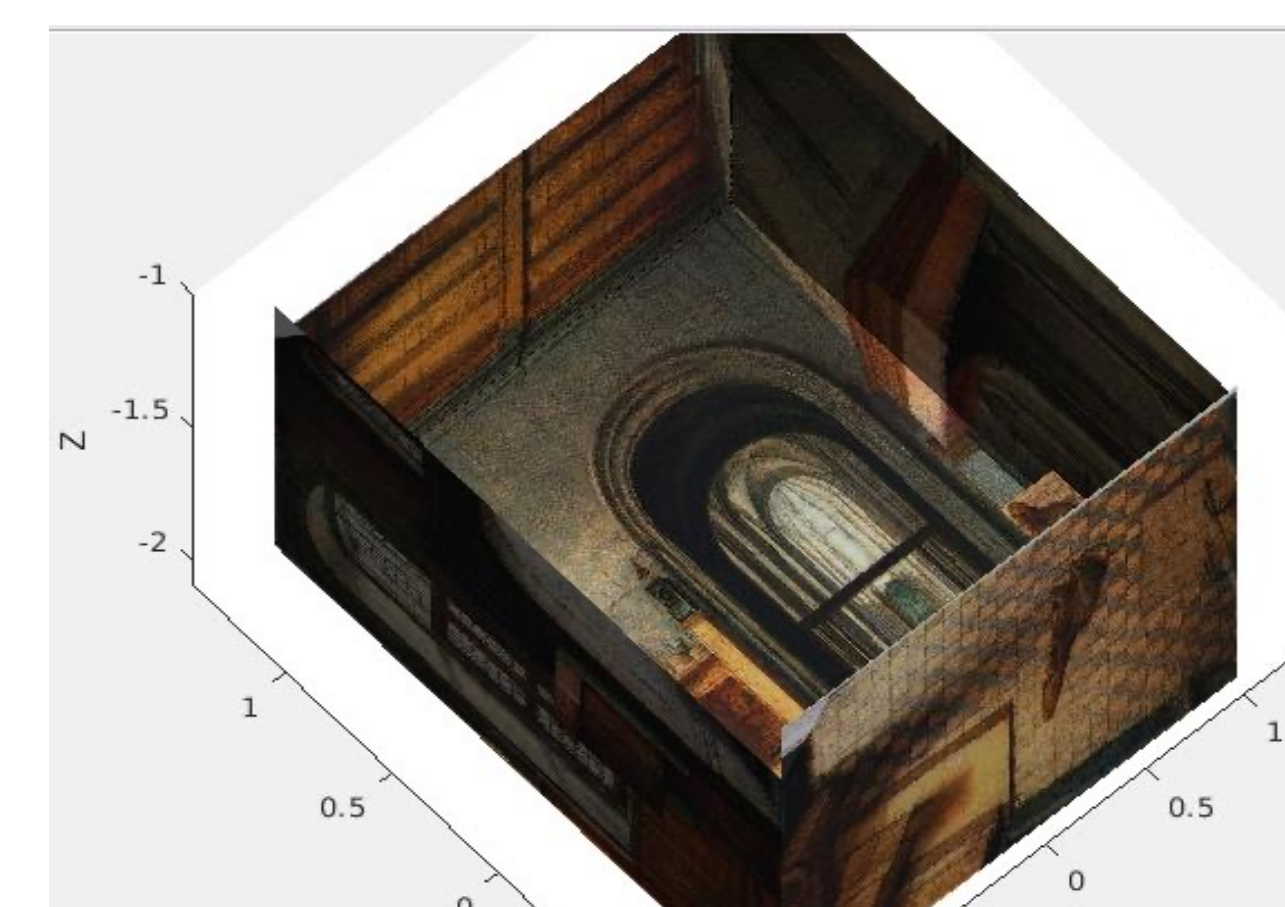
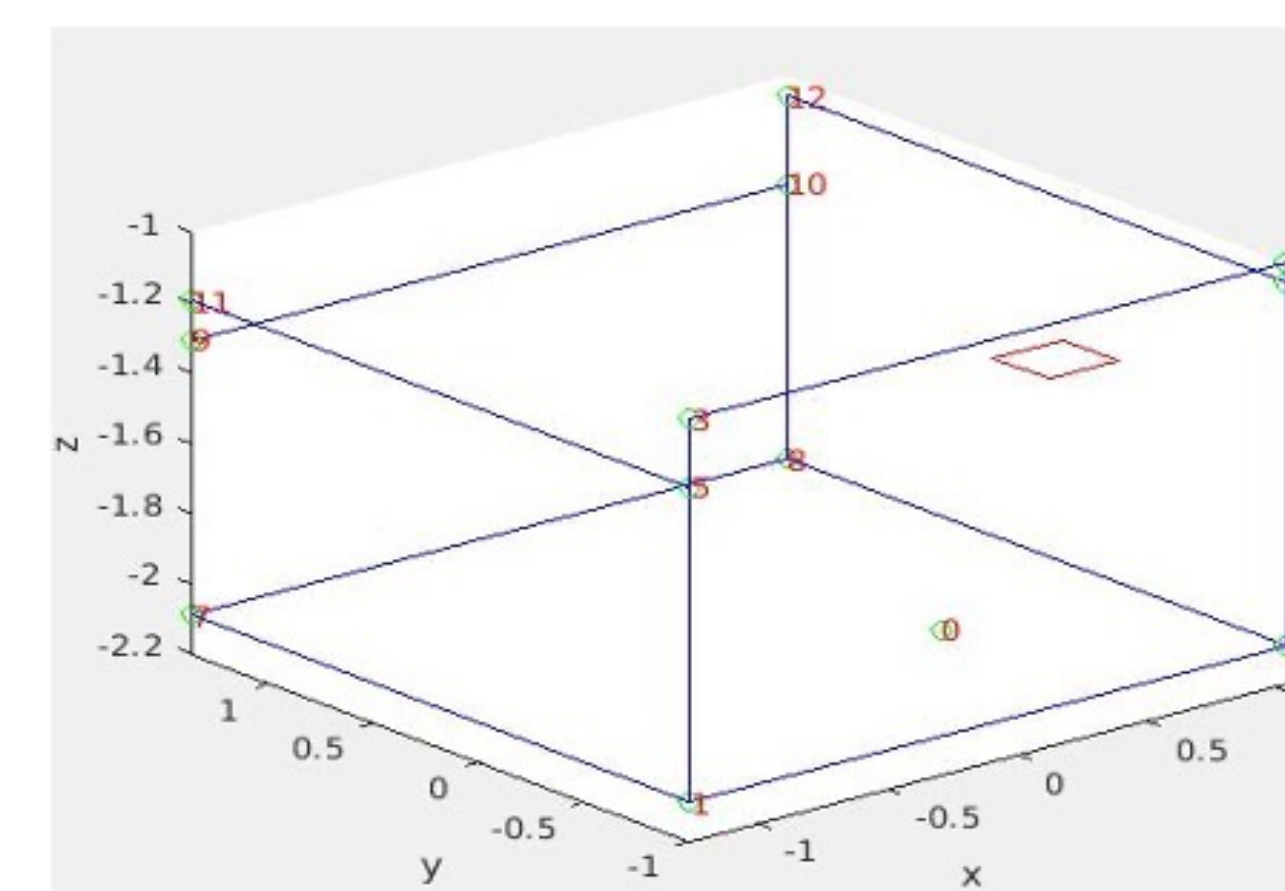
4 Generating spidery mesh

- Boundary of the 5 regions
- Intersections on the edge of the image



5 3D Reconstruction

- 2D pixel $\rightarrow$ 2D screen $\rightarrow$ 3D coordinate
- Selection attached background of foreground
- Perspective transformation of 5 subimages
- Mapping textures to meshes



6 Animation

- Change the camera position



Challenges

- Only one vanishing point
- Restriction of foreground modeling
- Medium execution speed without grayscale conversion

References

- [1]Horry Y, Anjo K I, Arai K. Tour into the picture: using a spidery mesh interface to make animation from a single image[C]//Proceedings of the 24th annual conference on Computer graphics and interactive techniques. 1997: 225-232.
- [2]Xinding Sun, J. Foote, D. Kimber and B. S. Manjunath, "Region of interest extraction and virtual camera control based on panoramic video capturing," in IEEE Transactions on Multimedia, vol. 7, no. 5, pp. 981-990, Oct. 2005, doi: 10.1109/TMM.2005.854388.
- [3]Deepu R, Murali S. 3D Reconstruction from Single 2D Image[J]. International Journal of Latest Research in Engineering and Technology (IJLRET), 2016, 2(1): 42-51.